Kian Kyars

San Francisco, CA kiankyars@gmail.com kiankyars.github.io kiankyars neuralkian

Python • PyTorch • TRL/vLLM • DeepSpeed • GRPO • Agent Evaluation

Experience

Judgment Labs — March 2026 - August 2026
Research Engineer
- Built a dynamic KNN+MST pipeline to cluster agent traces by failure mode (tool misuse, failed escalation) for root-cause analysis
- Built a self-improvement harness for software-engineering agents; ran 16+ hour unattended experiment loops
- Integrated a batched PII-redaction backend into the ingest pipeline, reducing cost 20×

Noemon — October 2025 - January 2026
Machine Learning Engineer
- Built a ReAct-based ARC-AGI-2 solver with tool-verified iteration and context compaction for multi-step agent workflows

fr8 Cohort 1.0 — June 2025 - August 2025
- Landed 35 merged PRs to Factorio Learning Environment across task-grounded agent evals, Gym infrastructure, and regression testing

Data-Driven Network and Cyber Security (DANS) Lab — September 2024 - June 2025
Undergraduate Research Assistant
- Fine-tuned Qwen2.5, Llama 3, and DeepSeek-R1 7-8B models and built local system-log anomaly detection matching GPT-4o on the evaluated task

MANGA NLP Lab — January 2025 - April 2025
Undergraduate Research Assistant
- Reimplemented the FLORA optimizer, improving 3B/7B LLM training speed up to 25% versus LoRA with AdamW

Hochschule für Technik und Wirtschaft Dresden — May 2023 - August 2023
Summer Research Intern
- DAAD RISE Intern; trained a 7-DOF Franka Emika arm in Nvidia Isaac Sim for block manipulation using TRPO and PPO

Selected Projects

Six Axes of an RL Environment — May 2026
- Designed 6 paired GRPO environments with TRL/vLLM on Qwen2.5-1.5B and ran 12 H100 experiments on Modal to isolate reward and tool-use failures

Reinforcement Learning from Verifiable Rewards (textbook) — April 2026
- Authored a 10-chapter reference on verifier design, reward hacking, training signals, and agentic RLVR

vLLM MoE Optimization (open PR) — July 2026
- Prototyped a fused Triton MoE combine reduction; measured up to 4.69× H100 microbenchmark speedup and added 33 parametrized test cases

CHIPS: From Design to Data Center — July 2026
- Published a source-checked course tracing an AI accelerator through design/EDA, fabrication, packaging, and data-center deployment

nanochat from Scratch — October 2025
- Reimplemented nanochat's tokenizer and NCCL-distributed training stack
- Additional: OpenClaw Course (250k+ views) • Pipeline Parallelism from Scratch • Hugging Face Ultra Scale Series • UofA Run Club • UofACarpool

Education

BSc Computer Science, Honours - University of Alberta — C/O 2025 *Grade: 3.9/4.0*